MIL-P-51104A(MU)
16 February 1970
SUPERSEDING
MIL-P-51104(MU)
21 January 1963

MILITARY SPECIFICATION

PROTECTOR, INFANT, M2

1. SCOPE

1.1 This specification covers one type of infant protector.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-375	-	Plastic Film, Flexible, Vinyl Chloride.
JJ-W-155	-	Webbing, Textile, (Cotton, Elastic).
CCC-D-950	-	Dyeing and Aftertreating Processes for Cotton Fabrics.
PPP-B-636	-	Box, Fiberboard.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing).

STANDARDS

FEDERAL

FED-STD-406	-	Plastics; Methods of Testing.
FED-STD-601	-	Rubber; Sampling and Testing.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

FSC 4240

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

C5-53-43	-	Diagram, Instruction.
DL5-53-62	-	Protector, Infant, M2.
DLB136-41-3000	-	Apparatus, Static Diffusion, Q170.
DLB136-42-102-8	-	Penetrometer, Filter Testing, DOP, Q127.

PUBLICATIONS

EDGEWOOD ARSENAL PURCHASE DESCRIPTIONS

197-51-350	-	Vinyl Chloride Polymer and Copolymer Flexible Sheets.
EA-F-1060	-	Filter Panel, Infant Protector.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publication forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM STANDARD

D747-63 - Stiffness of Plastics by Means of a Cantilever Beam.

3. REQUIREMENTS

3.1 Materials and components. All materials and components shall conform to the specifications and drawings forming a part of this specification.

3.1.1 Body and gussets. The material used for the body and gussets shall conform to type II, class 1 of L-P-375 except as specified in 3.1.1.1 through 3.1.1.5.

3.1.1.1 Stiffness. The stiffness requirement based on the Clark test shall be waived. The stiffness shall not exceed 6.0 X 10^4 PSI when tested as specified in 4.2.4.1.1.

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3.1.1.2 Volatility. Loss of weight shall not exceed 1.5 percent when tested as specified in 4.2.4.1.2.

3.1.1.3 Lacquer lifting. The lacquer-lifting requirement shall be waived.

3.1.1.4 Mustard resistance. Mustard resistance shall be a minimum of 30 minutes when tested as specified in 4.2.4.1.3.

3.1.1.5 GB resistance. GB resistance shall be a minimum of 78 minutes when tested as specified in 4.2.4.1.4.

3.1.2 Window. The material used for the window shall conform to 197-51-350 except as specified in 3.1.2.1 and 3.1.2.2.

3.1.2.1 Mustard resistance. Mustard resistance shall be a minimum of 30 minutes when tested as specified in 4.2.4.1.3.

3.1.2.2 GB resistance. GB resistance shall be a minimum of 125 minutes when tested as specified in 4.2.4.1.4.

3.1.3 Filter panels. Each filter panel shall conform to EA-F-1060.

3.1.4 Elastic webbing. The elastic webbing shall conform to JJ-W-155, except that piece dyeing will be permitted and the webbing shall be made mildew resistant in accordance with CCC-D-950, using type (a), (f), or (g) inhibitor.

3.2 Manufacture and assembly. The protector shall be manufactured and assembled as specified on Drawing D5-53-62 and subsidiary drawings referenced thereon.

3.2.1 Seam strength. The heat-sealed seams of the protector shall have a minimum strength of 17 pounds when tested as specified in 4.2.4.2.

3.2.2 DOP smoke penetration. DOP smoke penetration of the assembled protector shall not exceed 0.1 percent, when tested as specified in 4.2.4.3.

3.3 Workmanship. The protector shall be free from contamination (grease, oil, or foreign matter) and damage (cracks, holes, or tears).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

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4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Government's responsibility. The Government will be responsible for the performance of the tests in 4.2.4.1. Samples shall be forwarded to the laboratory designated by the contracting officer (see 6.2). Test specified in 4.2.4.2 will be observed by the Government.

4.1.3 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the protectors produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions. However, no more than one lot of filter matieral or one lot of vinyl material shall be represented in any one lot of finished protectors.

4.2.2 Sampling.

4.2.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For destructive tests.

4.2.2.2.1 Vinyl materials. A random sample, 10 square foot minimum, shall be taken from each lot of vinyl material and submitted to the Government for testing prior to the use of the material, from that lot, in the manufacture of protectors.

4.2.2.2.2 Seam strength. Twenty samples of each type seam (vinyl to vinyl, vinyl to window, and vinyl to filter panel assembly), ten for test and ten for retest, shall be produced from each lot of vinyl, window material, and filter panel plastisol, using the same materials, equipment, and methods intended for end item production.

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4.2.3 Inspection procedure.

4.2.3.1 For examination. Sample protectors shall be examined in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For destructive tests.

4.2.3.2.1 Vinyl material. Sample material shall be tested for compliance with the applicable requirement in 3.1.1 and 3.1.2.

4.2.3.2.2 Seam strength. Sample seams shall be tested for compliance with 3.2.1.

4.2.3.3 For critical defects. Each protector in each lot shall be inspected for critical defects (see 4.2.3.4).

4.2.3.4 Classification of defects.

(a) Protector, infant, M2 (Dwg D5-53-62).

Categories	Defects	Acceptance standards
Critical:		
1	DOP smoke penetration of assembled protector	4.2.4.3
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Marking incorrect	
103	Damage (3.3)	
104	Seams incorrect (width and continuity)	
105	Contamination (3.3)	
Minor:	AQL 2.5 percent defective	
201	Broken slats	
202	Marking missing or illegible	

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(b) Preparation for delivery (section 5).

Categories	Defects
Major:	AQL 6.5 percent defective
101	Container closure incorrect
102	Container damaged (crushed, torn, or punctured)
103	Instruction diagram missing or incorrect
104	Marking incorrect, missing, or illegible
Minor:	None defined

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 For vinyl materials.

4.2.4.1.1 Stiffness. Determine stiffness at 0°F in accordance with ASTM D747-63.

4.2.4.1.2 Volatility. Determine volatility in accordance with Method 6081 of FED-STD-406.

4.2.4.1.3 Mustard resistance. Determine the mustard resistance of each of five specimens with the Q170 Static Diffusion Apparatus (DLB136-41-3000). Take the average of the five determinations.

4.2.4.1.4 GB resistance. Determine the GB resistance of each of five specimens with the Q170 Static Diffusion Apparatus (DLB136-41-3000). Take the average of the five determinations.

4.2.4.2 Seam strength. Die out specimens from samples, at random locations, one from each type heat-sealed seam (vinyl to vinyl, vinyl to window, and vinyl to filter panel). Condition them for a minimum of 12 hours at 75° ± 3° F and a relative humidity of 50 ± 5 percent, and determine the seam strength in accordance with method 4311 of FED-STD-601. The lowest value obtained shall be considered as the seam strength. Breaking of the material at the edge of the seam may be considered as evidence of the seam strength.

4.2.4.3 DOP smoke penetration. The assembled protector shall be tested for DOP smoke penetration using the Q127 DOP Filter Testing Penetrometer (DLB136-42-102-8) at a flow rate of 7.5 liters per minute.

4.2.5 Acceptance/rejection criteria.

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4.2.5.1 Vinyl material. If any sample of vinyl material fails any requirement, another sample shall be tested to comply with the applicable requirement in 3.1.1 and 3.1.2. If the retest also fails, then the lot of vinyl material represented shall be rejected.

4.2.5.2 Seam strength. If any seam sample fails to comply with 3.2.1, then another like sample seam shall be tested. If the retest also fails, then the lot of protectors represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be as specified (see 6.2).

5.1.1 Method 1.

5.1.1.1 Infant protector sub-assembly (Dwg B5-53-62, less filter panels Dwg C5-53-44, see 6.3). Each sub-assembly shall be neatly folded and packaged in a fiberboard box conforming to style FOL, type CF, variety SW, class domestic, grade 275, of PPP-B-636. The box shall have approximate inside dimensions of 23-1/4 by 2-1/2 by 44-5/8 inches, length, width, and depth respectively. The box shall be closed in accordance with the closure requirements in the appendix to PPP-B-636, except that one end shall be sealed with 2-inch wide tape conforming to PPP-T-76. The tape shall run full length of the seams and extend over the ends not less than 2 1/2 inches.

5.1.1.2 Filter panels (Dwg C5-53-44). Filter panels shall be packaged in accordance with the preparation for delivery requirements in EA-F-1060.

5.1.2 Method 2. Infant protector assembly (Dwg D5-53-62). Each protector (see 6.4) shall be neatly folded and packaged in a fiberboard box conforming to style FOL, type CF, variety SW, class domestic, grade 275 of PPP-B-636. The box shall have approximate inside dimensions of 23 1/4 by 2 1/2 by 44 5/8 inches, length, width, and depth, respectively. The box shall be closed in accordance with the closure requirements in the appendix to PPP-B-636, except that one end shall be sealed with 2-inch wide tape conforming to PPP-T-76. The tape shall run full length of the seams and extend over the ends not less than 2 1/2 inches.

5.2 Marking. In addition to any special marking required by the contract or order, containers shall be marked in accordance with MIL-STD-129 and an instruction diagram conforming to Dwg C5-53-43 shall be applied to one side of the box by either printing, stenciling, or labeling. On the opposite side mark conspicuously "See Other Side for Instructions."

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6. NOTES

6.1 Intended use. This specification covers a protective shelter intended to be used to protect infants against chemical, bacteriological, and radiological warfare agents.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) The laboratory to which vinyl material samples should be sent for testing (see 4.2.2.2.1).
(c) Packaging method (see 5.1).

6.3 For maximum CK life of filter panels, the panels should be left in original sealed shipping container and should not be heat sealed into the protector until time of issue.

6.4 If the infant protectors, less filter panels, have previously been packaged in accordance with method 1, the tape onthe fiberboard box shall be cut so that no damage occurs to the box. The fiberboard box shall be re-used for packaging the assembled infant protectors.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project Number 4240-A374

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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